BRSTD: Bio-Inspired Remote Sensing Tiny Object Detection

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***Abstract*— In aerial images captured by drones or satellite remote sensing images, object information is weak and difficult to distinguish from the background, with significant variations in object sizes. Physiological research indicates that the visual system can select visual stimuli through an attention mechanism, focusing resources on processing important information while suppressing less important information. Inspired by biological vision, this article designs an object detection network, named bio-inspired remote sensing tiny object detection (BRSTD). Drawing inspiration from the parallel pathways in biological vision and the antagonistic receptive field properties of X, Y, and W cells, we designed the XYW-Conv module with antagonistic receptive fields. This module enhances the contrast between tiny objects and their surrounding information, effectively extracting tiny object information from images. To further improve the backbone network's ability to distinguish objects from the background, we designed the XYW-Attention and applied it to the designed Backbone. To better preserve tiny object information in the shallow feature layers and suppress large object and background information, we designed a feedback suppression attention module, top-down suppression attention (TDSA), at the connection between the neck and head parts, improving the model's performance. Experiments show that with only 1.8 M parameters, BRSTD achieved 10.4 in terms of AP_{vt} and 23.6 in terms of average precision (AP) on the AI-TOD dataset. It also performed excellently on other public remote sensing object datasets such as Visdrone and DOTA. This study not only advances remote sensing image object detection technology but also provides new ideas for the combined research of biological vision and computer vision (CV). The code will be open-sourced at https://github.com/huangyuesheng9/BRSTD.**

***Index Terms*— Aerial images, attention mechanism, bio-inspired, convolutional neural network (CNN), tiny object detection.**

I. Introduction

OBJECT detection [1], [2], a core area of computer vision (CV), stands alongside classification [3], [4], [5] and segmentation [6], [7] as one of the three principal tasks in the field. The primary tasks of object detection are object localization and classification, meaning that detectors must classify instances within an image and locate them using bounding boxes. From a task perspective, object detection serves as a bridge between classification and segmentation, underlining its significance. Traditional object detection methods typically follow a workflow of region selection, manual feature extraction, and classifier-based classification. However, due to the challenges in manually extracting diverse features, traditional algorithms have limitations in addressing object detection comprehensively. To overcome these constraints, recent years have seen the introduction of new methods and algorithms in the field, often based on deep learning technologies such as convolutional neural networks (CNNs). Compared to traditional approaches, deep learning-based object detection algorithms offer superior expressiveness and adaptability, learning features automatically from data and optimizing end-to-end on large datasets.

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In recent years, with the advancement of remote sensing technology, remote sensing images have been increasingly applied in both military and civilian domains, such as reconnaissance, terrain exploration, and air–ground coordination. Object detection is a crucial task in the field of remote sensing [8]. However, in UAV aerial images or satellite remote sensing images, tiny objects often occupy only a few dozen or even just a few pixels, as shown in Fig. 1(a). In addition, remote sensing tiny object detection datasets commonly have complex background information, as illustrated in Fig. 1(b). Dense object distribution, as shown in Fig. 1(c), is another frequent challenge. Moreover, the wide variety of target types and the significant scale variations among targets, as depicted in Fig. 1(d), further complicate detection tasks. These difficulties make tiny object detection in remote sensing a challenging aspect of object detection tasks.

To address these challenges, researchers have proposed various improvement measures: for example, increasing the resolution of input images [9] to ensure that tiny objects are not overlooked during feature extraction; fusing features from different levels [10] so that the model can simultaneously utilize low-level detail information and high-level semantic information, thereby improving the detection accuracy of tiny objects; and introducing background suppression techniques [11] to reduce the interference of background information on tiny object detection. These methods have improved the model's performance in detecting tiny objects to some extent, but several issues remain. Increasing the resolution of input images increases the model's complexity, leading to longer training and inference times, which is unfavorable for real-time detection in practical applications. Multilevel

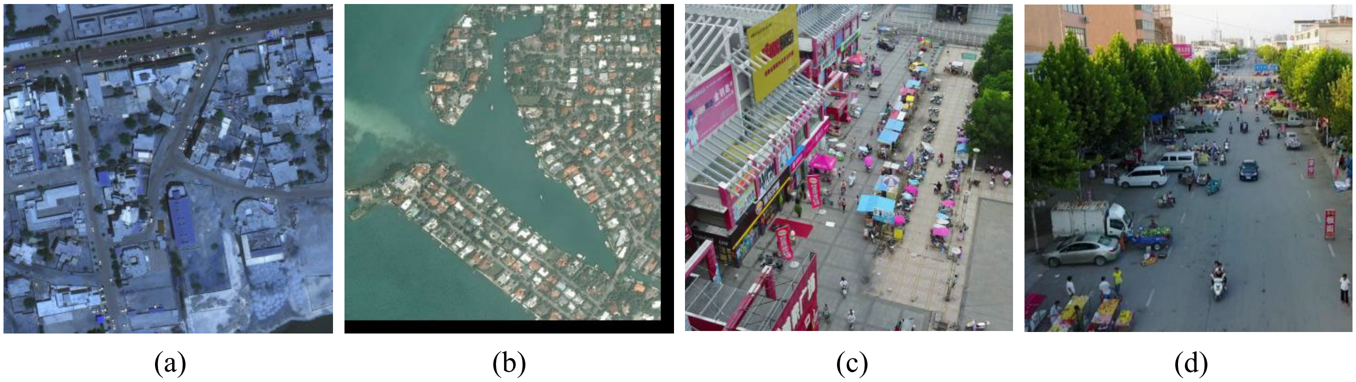


Fig. 1. Remote sensing image dataset is highly challenging. In (a), car objects are quite tiny in size. In (b), there is a significant amount of complex background information around the ships docked at the pier. In (c), distribution of bicycle is very dense. In (d), there are a wide variety of objects with substantial differences in their sizes.

feature fusion is difficult to optimize and increases the model's complexity. When suppressing complex backgrounds, there is a risk of mistakenly suppressing some tiny objects, resulting in missed detections. Moreover, these improvement measures largely rely on engineering approaches and computational techniques, lacking guidance from physiological theories.

Physiological studies have shown [12] that the human visual system extensively uses contextual information to facilitate object search in natural scenes. Katsuki and Constantinidis [13] theoretically explained that in top–down attention, the regulation of lower level information by higher level information can improve the accuracy of tiny object recognition. Inspired by biological visual attention mechanisms, Liao and Zhu [14] proposed YOLO-DRS, a bionic object detection algorithm for remote sensing images that integrates a multiscale, efficient, lightweight attention mechanism. Inspired by the human visual system's natural tendency to expend unequal energy on tasks with hierarchical difficulties for effective object recognition, Leng et al. [15] introduced a novel Pareto refocusing detection (PRDet). This network, under the guidance of inverse attention, differentiates between difficult and ordinary regions and refocuses on the difficult areas with the help of specific regional context. Beyond object detection, models inspired by biological vision have also garnered significant attention in other visual detection tasks such as contour detection [16], [17] and salience detection [18], [19]. Therefore, we will design two different attention networks, combining relevant physiological mechanisms to fully leverage the rich spatial and contextual information in images, as well as the hierarchical information within deep networks to address this challenge.

In summary, based on existing research achievements and related physiological evidence, we have designed a bio-inspired tiny object detection model inspired by biological vision mechanisms. By studying the receptive field properties of retinal ganglion cells, we provide a theoretical basis for the design of convolutional kernels that align with visual information processing mechanisms. By mimicking both bottom–up and top–down attention in biological vision, we introduce spatial channel attention with antagonistic receptive field properties, forming a top–down suppression attention (TDSA) that utilizes deep-level information to modulate lower level information, further enhancing the performance of remote sensing tiny object detection algorithms. This development not only advances object detection technology but also holds significant implications and potential applications in the intersection of computer and biological vision.

The contributions are summarized as follows.

1) Inspired by biological vision's parallel pathways and attention mechanisms, we propose a new lightweight tiny object detection model that achieves excellent performance, providing new insights for future researchers in machine vision.
2) Inspired by the parallel pathways and receptive field properties of XYW cells, we designed a new XYW parallel pathway convolution module (XYWConv), leading to a novel backbone that enhances the contrast between tiny objects and surrounding information, effectively boosting tiny object feature extraction capabilities.
3) Inspired by the bottom–up and top–down attention in the visual system, we constructed a bottom–up spatial channel attention (XYWA) based on the antagonistic receptive fields of X, Y, and W cells, enhancing the feature network's ability to distinguish between objects and the background. We also designed a top–down feedback suppression attention module to suppress large object and background information in shallow feature maps before the detection head, thereby enhancing tiny object features.

The remainder of this article is organized as follows. Section II discusses the related work concerning bio-inspired remote sensing tiny object detection (BRSTD). Section III provides a detailed introduction to our proposed methods. Section IV applies our methods to the AI-TOD [20] dataset and analyzes the performance and parameter size of the network in detail. We also present our experimental results on public datasets such as Visdrone [21] and DOTA [22], comparing them with other object detection methods. Section V concludes this article and discusses potential areas for further exploration.

II. Related Works

A. Object Detection

Object detection, as a core issue in CV, has consistently been a critical direction in the field. Traditional object detection algorithms, such as the Viola–Jones detector [23], [24], combine Haar-like features, integral images, Adaboost, and cascading classifiers to form a highly efficient and swift method. Its primary advantages are its speed and efficiency, though it struggles with accuracy in complex backgrounds. The deformable part model (DPM) [25] is another successful traditional detection algorithm that uses a histogram of oriented gradients (HOG) [26] and support vector machines (SVMs) [27] to build gradient-based models of objects, and DPM excels at handling partial occlusions but suffers from high computational costs, making it unsuitable for real-time applications.

In recent years, fueled by extensive research efforts, object detection technologies based on deep learning have rapidly advanced, continuously improving in both speed and accuracy. Depending on the steps required in the detection process, detectors can generally be divided into two categories: two-stage detectors and one-stage detectors. Two-stage detectors first generate proposal regions using a region proposal network (RPN), followed by localization and classification of objects within these proposed areas, as seen in Faster R-CNN [28], Cascade R-CNN [29], and TridentNet [30]. Redmon et al. [31] introduced you only look once (YOLO), the first one-stage detector in the deep learning era, which operates at extreme speeds up to 155 frames/s, surpassing all two-stage detectors of the time, and has since evolved through several iterations [32], [33], [34], [35], [36]. In the same year, the single-shot multibox detector (SSD) was proposed by Liu et al. [37], incorporating multireference and multiresolution detection techniques, significantly enhancing the accuracy of one-stage detectors, especially for tiny objects. Cornernet [38], the first anchor-free detector, eliminates the need for predefined anchor boxes, thereby avoiding complex computations related to anchors and all associated hyperparameters, ushering in a new era for one-stage detectors. As one-stage detectors can directly classify and localize semantic objects on feature maps through dense sampling without the need for RPN, they typically have fewer parameters and faster detection speeds. With the rapid progress of deep CNNs, one-stage detectors have surpassed two-stage detectors in terms of real-time performance and simplicity of design.

B. Tiny Object Detection in Remote Sensing

The detection of tiny objects in remote sensing has long been a challenge in the field of object detection due to characteristics such as low pixel ratio of the objects, limited information content, and high requirements for localization accuracy.

Research [39] has shown that appropriate modeling of context within the backbone can enhance object detection performance, especially for remote sensing tiny objects, which have inconspicuous appearance features. Li et al. [40] introduced a single detector-based feature fusion (SDFF) architecture in HBD-Net, which incorporates multiscale context into a dynamic image pyramid (DIP) for large-size images through interlayer feature fusion (IFF) to address the issue of significant object size variation in remote sensing images. Zheng et al. [41] proposed a reinforced context model network that enhances shallow context information by expanding the effective receptive field, effectively improving the feature expression capability of tiny objects. Chen et al. [42] introduced a context fine-tuning network, which first identifies context areas related to the object area by computing similarities and then enhances the object area features using these context area features. Although this strategy sacrifices some processing speed, it significantly enhances the capture of relevant context. Ruan et al. [43] introduced the multireceptive field and detail mining (MRF-DM) module to fuse multiscale receptive fields for learning higher level semantic representations. Feng et al. [44] designed a global context-aware enhancement (GCAE) module that activates features of the entire object by capturing the global visual scene context. Wu et al. [45] designed a feature refinement module (FRM), which refines features by convolving multiple receptive fields across different branches, thus improving feature discrimination at various scales. The aforementioned works utilize multiscale receptive fields to fully leverage the rich context information in images, thereby enhancing the performance of tiny object detection, but their algorithmic implementations are relatively complex and require high computational resources.

Attention mechanisms play a crucial role in enhancing the ability of neural networks to represent tiny objects, inspired by human perception of visual information. Scholars have proposed various attention computation models, which are now widely applied in remote sensing object detection tasks. Lim et al. [46] proposed object detection with an attention mechanism, which can focus on objects in the image and include context information from the target layer. Ran et al. [47] proposed a multiscale context and enhanced channel attention (MSCCA) model, which enhances feature map channel attention and integrates multiscale context information with multiscale detection methods, improving the CNN's ability to represent remote sensing tiny objects. Li et al. [48] designed a cross-layer attention module to capture nonlocal associations of tiny objects at each layer and further enhance their representational power through cross-layer integration and balancing. Dong et al. [49] introduced several attention-based multilevel feature fusion modules (A-MLFFMs) in the neck section to further implement multiscale object detection by integrating the outputs of the FPN. Qu et al. [50] proposed a remote sensing tiny object detection network based on attention mechanisms and multiscale feature fusion, which strengthens the representation of remote sensing tiny object features through the combination of multiscale feature fusion and attention mechanisms. Inspired by the attention mechanisms in these CNNs and further combining with the current mainstream theories of attention mechanisms in biological vision, we propose two different types of attention mechanisms: the XYWA, inspired

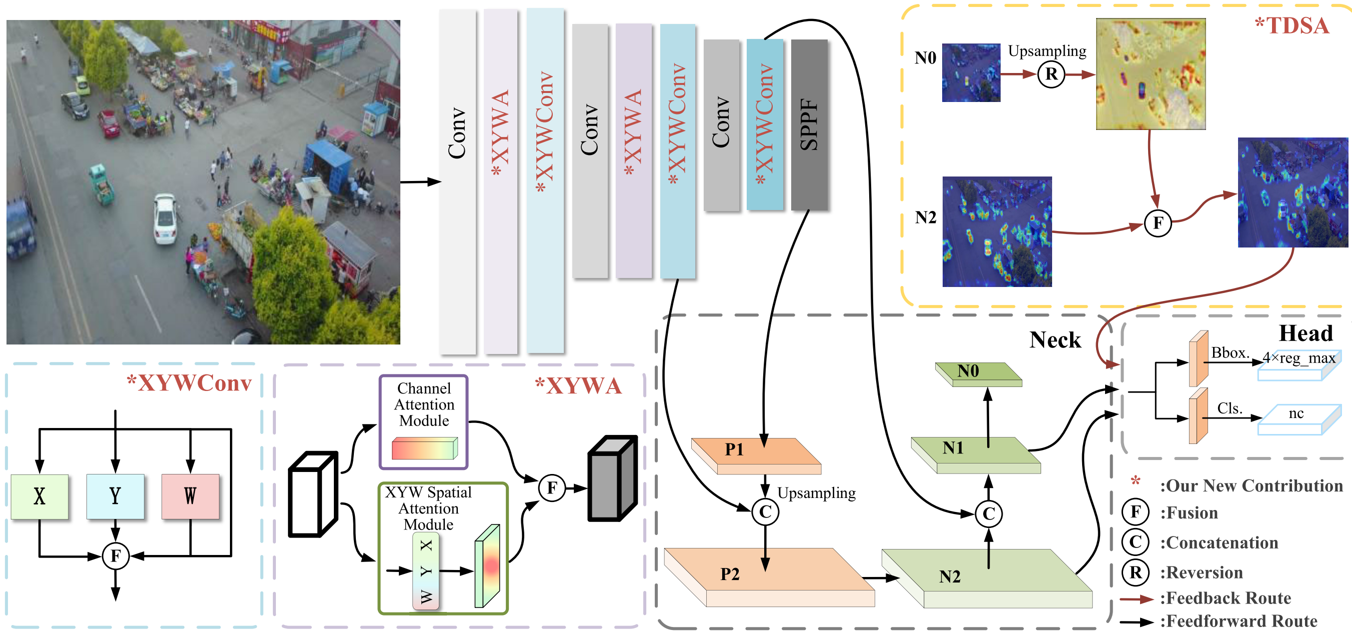


Fig. 2. Overview of the proposed BRSTD framework. Our main contributions include: 1) XYWConv inspired by the parallel visual pathways and XYW cell receptive fields; 2) bottom–up multiscale attention XYWA composed of XYWConv; and 3) feedback suppression attention TDSA inspired by top–down attention in biological vision. After preprocessing the input image with techniques such as mosaic, it is fed into the backbone network composed of XYWConv and XYWA and then into the neck network composed of C2f. Unlike the neck in Ultralytics, our neck network has one less layer, and N1 is directly connected laterally with the backbone feature layer of the same size instead of P1, aiming to retain as much detail of small objects in the image as possible. N1, after downsampling, becomes N0 and is sent into TDSA along with N2. Subsequently, the output of TDSA, along with N1 and N2, is sent to the head for detection.

by the bottom–up attention in biological vision, and the TDSA, inspired by the top–down attention.

III. Method

We propose a novel one-stage object detection network with a backbone-neck-head architecture, as depicted in Fig. 2. In the design of the backbone, we incorporate the XYWConv module, inspired by the receptive field properties of XYW cells, to extract features, with detailed discussions in Section III-A. In addition, we introduce our uniquely designed XYW spatial channel attention (XYWA) to enhance the network's ability to discriminate between target objects and the background, with detailed discussions in III-B. Furthermore, in the head part of the network, we design a novel suppression attention module for tiny object detection, named TDSA, inspired by top–down attention mechanisms. This module aims to suppress information about larger objects in shallow feature layers, thereby accentuating tiny object information, which will be elaborately discussed in Section III-C.

A. Bio-Inspired Feature Extraction Module

Research shows that neurons in the lateral geniculate nucleus (LGN) get feature information from retinal ganglion cells through different channels. X-type LGN neurons process information from X-type ganglion cells, Y-type from Y-type, and W-type from W-type ganglion cells [51]. Once processed, these neurons send the details to specific cells in the visual cortex [52]. The X pathway involves small, closely packed cells with tiny receptive fields, great for spotting fine details and color differences in small objects, essential for identifying them against complex backgrounds [53]. The Y pathway has cells with bigger receptive fields, capturing broader contextual information from images and sensitive to light changes, highlighting features of small objects in varied lighting [54].

The W pathway helps control tiny eye movements, refining the eye's focus on targets to enhance contrast and detail perception [55].

In the task of detecting tiny objects in static images, the cooperation of these three pathways ensures effective recognition and localization of objects from various angles and scales. The X pathway provides the necessary detail information, the Y pathway enhances the visual prominence of the objects, and the W pathway contributes to background contrast, collectively promoting efficient and accurate tiny object detection. Inspired by these physiological mechanisms [51], [52], [53], [54], [55], we developed three pathway modules, XYW, and further assembled into a new XYWConv module. The specific structure of this module is shown in Fig. 3.

In the X cell pathway, X cells have a unique receptive field that follows the Rodieck model [53], featuring a small central area sensitive to light and a larger surrounding area that reacts oppositely. This setup, shown in the lower left of Fig. 4(a) with a green line for the center and a blue line for the periphery, allows X cells to highlight fine details such as small objects and edges by enhancing contrasts caused by uneven lighting. As depicted in Fig. 4(a), X cells are excellent at boosting the visibility of small features and edges against complex backgrounds.

To mimic this natural capability in technology, we designed a convolutional module based on the X cell's receptive field characteristics. Unlike traditional methods using the difference of Gaussians [53], our approach uses learnable convolutional

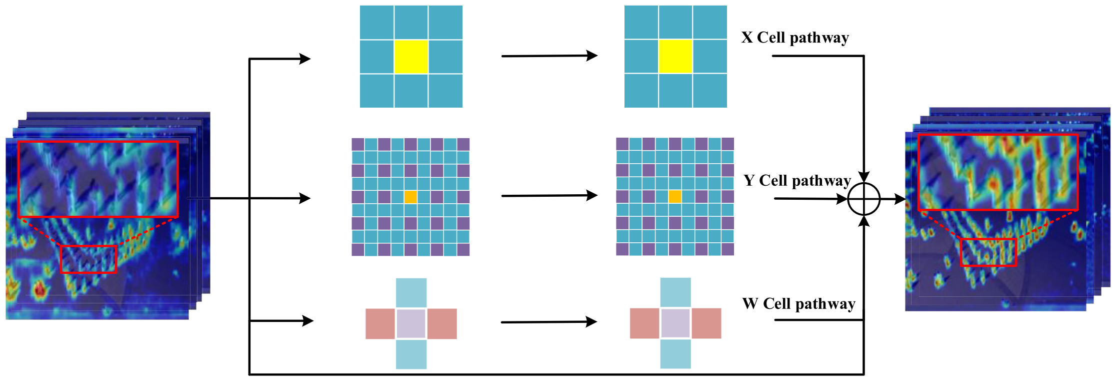


Fig. 3. XYWConv inspired by parallel pathways. It is composed of a parallel X cell pathway, a Y cell pathway, and a W cell pathway. To mitigate the issues of gradient vanishing and gradient explosion, residual connections are incorporated into the module design. From the visualization results of the input and output heatmaps, it can be seen that after processing with XYWConv, the feature range of tiny objects becomes clearer, and the distinction from the surrounding background is significantly improved.

kernels. Specifically, we use a 1×1 kernel for central features and a 3×3 kernel for peripheral information and then subtract the peripheral from the central to replicate the center-surround antagonism. This design is detailed in the upper left of Fig. 4(a), showing how our model effectively simulates the natural response of X cells.

In addition to the classical center-surround setup, Y cells have a more complex structure featuring numerous nonlinear subunits with rectifying properties [56], [57], as depicted by the red dashed line in the lower left of Fig. 4(b). These subunits, though small, are scattered throughout Y cells' receptive fields, functioning similar to the central mechanism and showing sensitivity to light [53]. They are especially crucial in picking up tiny objects by amplifying sudden changes in an image, particularly high-frequency details. Moreover, these subunits can enhance the detection of specific spatial frequencies, making them valuable for identifying tiny objects of certain sizes and shapes [56], [57], [58].

Building on these insights, we developed a convolutional module that replicates Y cells' receptive field characteristics, showcased in the upper left of Fig. 4(b). Considering that Y cells generally have larger receptive fields than X cells, we use a 9×9 kernel for peripheral simulation. For the central response, we continue using a 1×1 kernel. To imitate the unique nonlinear subunits of Y cells, we use a dilation rate of 2 and a 5×5 kernel size, matching the effective field size of the 9×9 peripheral convolution, allowing our design to closely mimic the complex structure of Y cells' receptive fields.

W cells, discovered after X and Y cells, have the function of controlling micromovements of the eyeballs [55], and the receptive field model of W cells is shown on the left in Fig. 4(c). These micromovements enable the eyes to gather some peripheral information while focusing on a specific point. In this article, a cross-shaped convolution composed of 1×3 and 3×1 convolutions is used to simulate the receptive field of W cells, as shown on the right in Fig. 4(c).

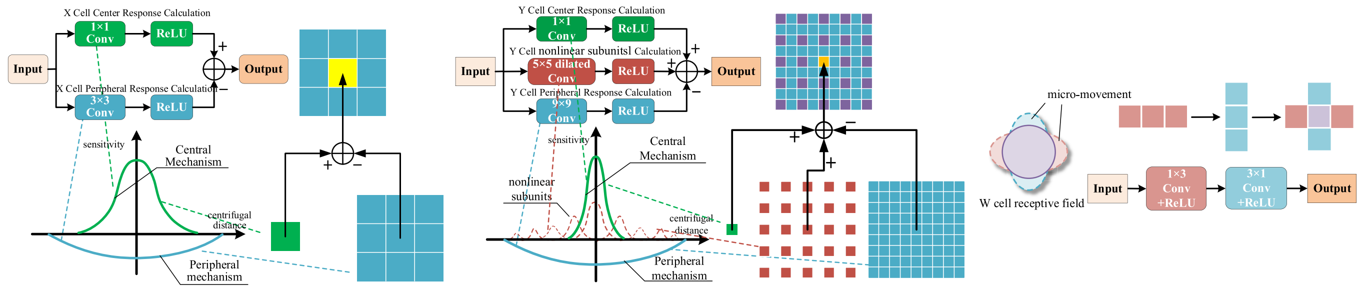
B. Bio-Inspired Spatial Channel Attention

There are two main theoretical models for bottom–up spatial attention: the parietal salience model by Itti et al. [59] and the V1 salience model by Li [60], [61]. The V1 model highlights that V1 neuron connections enhance sensitivity to context, creating a salience map. The Mexican hat model [62], [63] focuses on neural responses in the retina, showing a narrow inhibitory region surrounded by a wider activation area, which looks like a Mexican hat. This pattern boosts the contrast between focused and unfocused areas. Unlike spatial attention, feature-based attention is spatially global, allowing dynamic reweighting of various feature maps to create global changes in feature processing [63], [64], [65]. Hayden and Gallant [66] believe that the effects of spatial and feature attention appear to be additive.

Inspired by these models, we have developed a new attention mechanism for our backbone system. It uses the spatial attention from the retina to the V1 area via XYW parallel pathways [51]. It introduces XYWConv in XYWA for generating spatial attention weights using a mix of multiscale kernels and convolutions. Similarly, emphasizing a center-surround inhibitory effect, the Mexican hat model informs our X and Y convolutions, enhancing sensitivity to spatial contrasts. Betts et al. [67] observed that this antagonistic mechanism's effectiveness declines with age and is particularly impactful for detecting small, low-contrast objects. Thus, it is especially suitable for tiny object detection tasks. In addition, we have designed a channel attention mechanism that adjusts the importance of different features by dynamically weighting the channel level.

As illustrated in Fig. 5, the structure of XYWA is composed of two parts: spatial attention and channel attention. The core of the spatial attention is the previously discussed XYW cell pathway modules. These modules utilize XYW channels with antagonistic mechanisms to capture multiscale contextual information, which are then combined to form XYW-spatial attention weights. On the other hand, global average pooling compresses the input feature maps along the H and W dimensions, and channel-wise attention weights are obtained through two linear transformations.

After performing matrix multiplication between the two types of weights, the resulting matrix is element-wise multiplied with the original input features to highlight important information and suppress the less relevant details. Finally, a residual connection is used between the input and output



(a) (b) (c)

Fig. 4. X, Y, and W convolution module inspired by X, Y, and W cell receptive fields. (a) X cell antagonistic receptive field mode demonstrates the XConv module's computation process using convolution kernels. (b) Y cell antagonistic receptive field nonlinear model demonstrates the YConv module's computation process using convolution kernels. (c) W cell flicker receptive field model demonstrates the WConv module's computation process using convolution kernels.

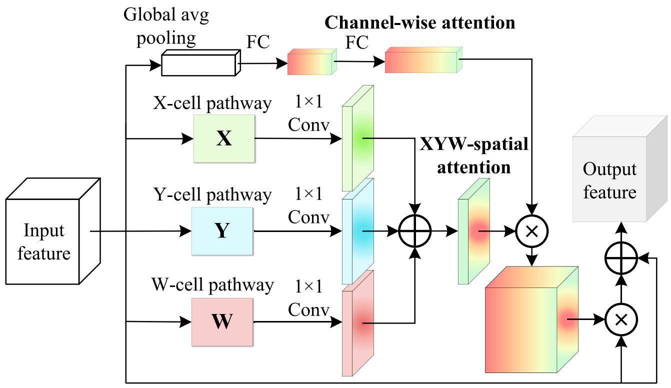


Fig. 5. XYW spatial channel attention module. Using parallel X, Y, and W cell pathways to generate multiscale spatial attention weights. Channel attention weights are produced through average pooling and fully connected operations. Multiplying the spatial and channel attention weights yields the spatial channel attention weights.

to enhance the output information while addressing issues of gradient vanishing and explosion. The overall process can be expressed as follows:

$$F_{\text{out}} = F_{\text{in}} \odot W_A + F_{\text{in}} \tag{1}$$

where F_{out} is the output feature map, F_{in} is the input feature map, and W_A are the weights from the spatial channel attention, sized $C \times H \times W$. The operator $\odot$ denotes element-wise multiplication.

The computation formula for the weights W_A can be expressed as

$$W_A = \text{sigmoid}(W_c \odot W_s). \tag{2}$$

In the structure, W_c represents the channel attention weight with dimensions $C \times 1 \times 1$, and W_s is the spatial channel attention weight with dimensions $1 \times H \times W$. These weights are combined through element-wise multiplication with broadcasting, and the final weights W_A are obtained after applying a sigmoid activation function.

The calculation formula for the spatial attention weight W_s with antagonistic mechanism is as follows:

$$W_x = C_x^{1\times1}(f_x(F_{\text{in}})) \tag{3}$$

$$W_y = C_y^{1\times1}(f_y(F_{\text{in}})) \tag{4}$$

$$W_w = C_w^{1\times1}(f_w(F_{\text{in}})) \tag{5}$$

$$W_S = W_x + W_y + W_w \tag{6}$$

where f_x, f_y, and f_w represent the XYW pathway modules discussed in Section III-A. Each module processes the input feature map F_{in} and outputs to 1×1 convolutional kernels $C_x^{1\times1}$, $C_y^{1\times1}$, and $C_w^{1\times1}$, which reduces the channel count to one. This results in the weights W_x, W_y, and W_w each having dimensions $1 \times H \times W$. Summing these weights produces W_S.

C. Bio-Inspired Feedback Inhibition Attention

In deep learning networks, shallow feature layers often contain more detailed information, while deeper layers contain more compressed semantic information. Semantic information can be considered as high-level information produced by advanced associative cortices of the brain, including the prefrontal cortex (PFC) and the posterior parietal cortex (PPC), whereas detail information is seen as low-level information generated by lower visual cortices (V1 and V2). Moore and Fallah [68], [69] demonstrated through experiments that the PFC is involved in regulating attention mechanisms. Ekstrom et al. [70] provided direct evidence for top–down attention mechanisms through their experimental findings. Katsuki and Constantinidis [13] elaborated on the principle in which high-level information regulates low-level information in top–down attention.

As networks deepen, many details of tiny objects are lost during subsampling, while information about larger objects remains more intact. Furthermore, in the detection process of shallow feature layers, tiny object features can be overshadowed by the stronger features of larger objects. To address this issue, inspired by the characteristics of top–down attention mechanisms, we designed a suppression attention module TDSA between the neck and head sections of the network. This module utilizes deep-level information to modulate low-level information, enhancing weak features of tiny objects in shallow feature layers and suppressing features of medium to large objects. The internal structure of this module is shown in Fig. 6.

F_s and F_d are the input features for the TDSA, where F_s represents a shallow large-scale feature map and F_d represents a deep small-scale feature map, with dimensions $C_s \times H_s \times W_s$ and $C_d \times H_d \times W_d$, respectively. Here, $C_s < C_d$ and $H_s, W_s > H_d, W_d$. Thus, F_d first requires transformations in both channel and spatial dimensions before

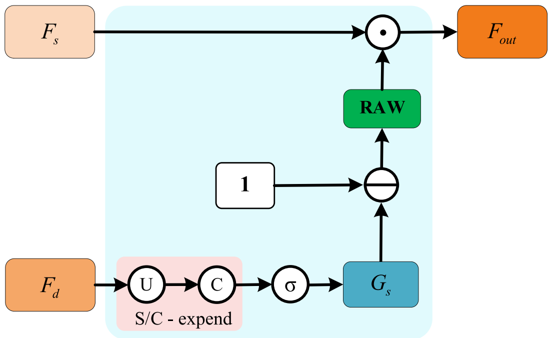


Fig. 6. Using top–down feedback suppression attention to inhibit interference information in shallow feature maps with deep feature information.

further processing. The transformation formula is

$$G_s = \text{sigmoid}(C(U(F_d))). \tag{7}$$

Here, U represents an upsampling function, used to elevate the spatial dimensions H_d and W_d of F_d to match those of H_s and W_s of F_s. C denotes a 1×1 convolutional kernel, utilized for scaling in the channel dimension. The final weights G_s are obtained after applying a sigmoid activation function. Due to the broadcasting mechanism of tensor operations, the inhibition attention weight is obtained as

$$\text{RAW} = 1 - G_s. \tag{8}$$

Finally, the input F_s is element-wise multiplied with RAW to produce the output F_{out}. The TDSA effect is shown in Fig. 7.

IV. Experiments

This experiment utilized the AI-TODv2 and VisDrone2019 remote sensing weak tiny object datasets to validate the detection performance of the model. The model was trained using the Ultralytics [71] framework on an NVIDIA RTX 4090 GPU for 550 epochs, with the batch size set to 4. During training, data augmentation techniques, such as mosaic augmentation, flipping, scaling, and color space transformations, were employed. Stochastic gradient descent (SGD) was used as the optimizer for all models, with a learning rate (lr) of 0.01 and a momentum set to 0.937. The lr warmup was set to three epochs, and lr decay was implemented using the CosineAnnealing algorithm. Our designed model did not use any pretrained weights throughout the training process.

A. Evaluation Metrics

This article primarily uses average precision (AP), mean AP (mAP), and the number of parameters as the standards to measure the model performance. The formulas for calculating AP and mAP are as follows:

$$\text{AP} = \int_0^1 P\,dR \tag{9}$$

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i. \tag{10}$$

AP represents the integral of precision (P) over recall (R), with confidence thresholds ranging from 0 to 1. mAP represents the mAP value across all categories in the dataset, where N is the number of categories. Higher values of AP and mAP indicate stronger detection performance of the model. The formulas for calculating precision and recall are as follows, where TP, FP, and FN denote true positives, false positives, and false negatives, respectively:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{12}$$

Specifically, we have categorized AP and mAP into AP50, mAP50, AP50-95, and mAP50-95. AP50 and mAP50 refer to the values of AP and mAP when the IoU threshold defining a true positive is set at 0.5. AP50-95 and mAP50-95 represent the average values of AP and mAP obtained when the IoU threshold ranges from 0.5 to 0.95, with an interval of 0.05. These metrics are currently the most widely used evaluation standards in the field of object detection.

While AP and mAP are similar in their calculation methods, they differ in specific applications and interpretations. In the Ultralytics framework, the experimental results are often presented in the form of mAP, with the official evaluation metric for the DOTA dataset being mAP50. However, for datasets such as AI-TOD, there are strict requirements for evaluation metrics, such as AP_{vt} and AP_t. Therefore, many researchers use the AP metric in control experiments for the AI-TOD dataset. To compare with other state-of-the-art (SOTA) models, we use the most common evaluation metrics in each dataset for experimental analysis. Specifically, we use the AP evaluation metric in the AI-TOD and VisDrone datasets, and the mAP evaluation metric in the DOTA dataset. In ablation studies, we only need to experiment and compare based on our own model without comparison to other models, making the use of the mAP metric under the Ultralytics framework more convincing.

B. Datasets

We validate our model on three aerial weak tiny object datasets: AI-TODv2 [20], VisDrone2019 [21], and DOTA [22].

AI-TODv2 is an extremely challenging dataset designed specifically for the detection of weak tiny objects. It comprises 28 036 aerial images of 800 × 800 pixels, including 11 214 images in the "train" set, 2804 images in the "val" set, and 14 018 images in the "test" set. The dataset covers eight categories, including airplane, bridge, and storage tank, with a total of 700 621 object instances. The average object size is only 12.8 pixels, which is much smaller compared to other existing aerial object detection datasets.

The **VisDrone2019** dataset contains 10 209 aerial images, including 6471 images in the "train" set, 548 images in the "val" set, and 3190 images in the "test" set. It includes categories such as pedestrian, people, and bicycle, totaling ten different categories. It also includes a large number of densely distributed tiny objects, making the VisDrone dataset particularly challenging.

The **DOTA** dataset, introduced in 2018, is a highly authoritative dataset in the field of remote sensing object detection. It is divided into 15 categories, containing a total

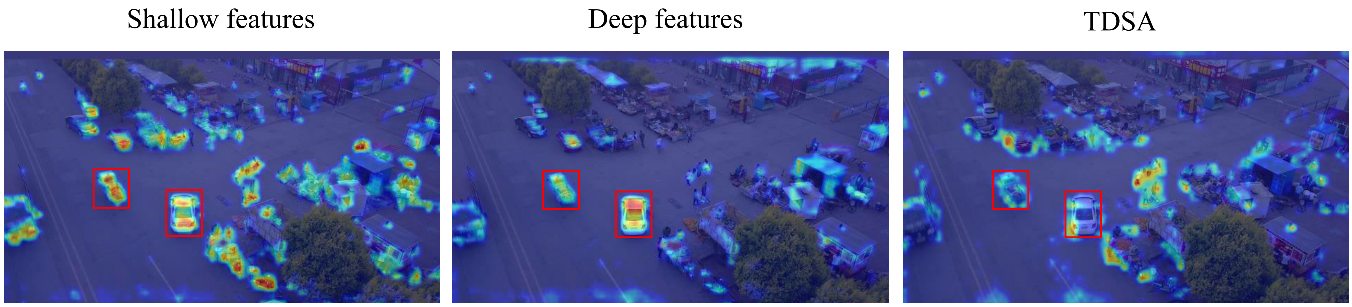


Fig. 7. After TDSA, the large object information in the shallow feature layer is suppressed.

TABLE I
QUALITATIVE RESULT OF THE AI-TOD TEST SET. NOTE THAT RED REFERS TO THE BEST RESULTS AND BLUE REFERS TO THE SECOND-BEST RESULT. * INDICATES THAT THE MODEL IS BASED ON TRANSFORMER

Model	Venue	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m	Param/M	GFLOPs
Faster RCNN[28]	TPAMI2017	11.1	26.3	7.6	0.0	7.2	23.3	33.6	236.3	289.3
Cascade RCNN[29]	CVPR2018	13.8	30.8	10.5	0.0	10.5	25.5	36.6	319.4	146.6
DETR*[72]	ECCV2020	2.7	10.3	0.7	0.7	2.1	3.0	12.4	41	225
ATSS[73]	CVPR2021	12.8	30.6	8.5	1.9	11.6	19.5	29.2	244.5	-
CenterNetV2[74]	Arxiv2021	14.1	35.7	8.5	2.9	12.7	19.7	29.5	44.9	206
DetectoRS[75]	CVPR2021	14.8	32.8	11.4	0.0	10.8	28.3	38.0	540.1	246
Deformable-DETR*[76]	ICLR2021	17.0	45.9	8.8	7.2	17.1	22.7	28.2	40	196
DAB-Deformable-DETR*[77]	ICLR2022	16.5	42.6	9.9	7.9	15.2	23.8	31.9	44	256
FSANet[78]	TGRS2022	20.3	48.1	14.0	6.3	19.0	26.8	36.7	31.9	125.2
NWD-RKA[79]	ISPRS2022	23.4	53.5	16.8	8.7	23.8	28.5	36.0	540.1	246
RFLA[80]	ECCV2022	24.8	55.2	18.5	9.3	24.8	30.3	38.2	-	-
YOLOv8-m[71]	Arxiv2023	22.9	52.5	16.5	7.1	22.4	29.7	37.1	25.8	78.7
HANet[81]	TCSVT2023	22.1	53.7	14.4	10.9	22.2	27.3	36.8	26.4	-
DINO-Deformable-DETR*[77]	ICLR2023	23.2	56.6	15.4	9.9	23.1	29.3	37.6	44.0	279
KLDet[82]	TGRS2024	19.6	-	-	8.4	20.6	22.7	26.4	32.0	-
MENet-Swin-T*[83]	TGRS2024	23.2	56.2	15.0	9.7	23.9	25.3	34.4	41.9	148.3
ORFENet[84]	TGRS2024	24.8	55.4	18.2	9.7	24.4	28.7	35.1	32.7	373.3
BRSTD(our)	-	23.6	54.2	16.6	10.4	22.8	30.3	37.6	1.8	64.3
BRSTD_l(our)	-	26.1	58.0	19.5	11.6	26	32.3	38.5	6.3	220.2

of 2806 large images and 188 282 instances. Each original image is 4000 × 4000 pixels in size. For experimental use, these images are cropped into 1024 × 1024 pixels without any overlap, totaling 25 254 images. Of these, 15 749 images are designated for the "train" set, 5297 images for the "val" set, and 4208 images for the "test" set. The input size for the images fed into the network is fixed at 640 × 640 pixels.

C. Comparisons With the SOTA Methods

We proposed two versions of our network model for comparative and ablation studies: lightweight version BRSTD and its enhanced version, BRSTD-l. The primary difference between the two is the width of the network, and BRSTD-l has double the number of channels in each layer compared to BRSTD. In the comparative studies, these models were benchmarked against classic object detection models as well as current SOTA models.

1) Experiment on AI-TOD: In the AI-TODv2 dataset, we employed the most widely used method for training and testing, which involves using the "train + val" sets for training and the "test" set for evaluation. For performance comparison with other models, and to benchmark against current SOTA models, we utilized pycocotools for testing. Specifically, according to [20], AP_{vt}, AP_t, AP_s, and AP_m represent the AP for very tiny (2 pixels), tiny (8–16 pixels), small (16–32 pixels), and medium (32–64 pixels) scales, respectively. The results of the performance comparisons are presented in Table I.

In Table I, we compare the performance of two versions of BRSTD with other strong baselines on the AI-TOD test set, which includes leading models in CV tasks [28], [29], [72], [74], [75] as well as SOTA models for remote sensing images [78], [79], [81], [82], [83], [84]. Overall, our BRSTD_l outperforms the latest remote sensing object model

ORFENet [84] (AP_{vt} = 9.7, AP_t = 24.4, and Params = 32.7) by at least 1.9 AP_{vt} (19.6%) and 1.6 AP_t (6.6%) while reducing the parameter count by 26.4 M (80.7%). Compared to the second-best comprehensive performance model RFLA [80] (AP_{vt} = 9.3 and AP_t = 24.8), our model also achieves an improvement of 2.3 AP_{vt} (24.7%) and 1.2 AP_t (4.8%) and surpasses it in other metrics as well. The lightweight BRSTD model, with only 1.8 M parameters, also outperforms ORFENet and RFLA in detecting very tiny objects, by 0.7 AP_{vt} (7.2%) and 1.1 AP_t (11.8%), respectively. BRSTD also reaches 95.1% and 97.8% of ORFENet's performance in terms of AP and AP_{50} metrics.

As shown in Table I, the BRSTD_l model demonstrates significant advantages over other SOTA models in both detection performance and parameter count, particularly excelling in parameter efficiency, second only to the lightweight BRSTD model. Despite its extremely low parameter count, the BRSTD model achieves excellent performance. Compared to the classic lightweight model YOLOv8-m [71], BRSTD has only 7% of its parameter count, yet achieves an improvement of 3.3 AP_{vt}. In addition, BRSTD's computational complexity is 14.4 GFLOPs lower than that of YOLOv8-m. It is worth noting that under the same experimental conditions, BRSTD reaches an FPS of 143.5 (runtime 0.0069 s), surpassing YOLOv8-m's 138 frames/s (runtime 0.0072 s), which fully demonstrates the high efficiency of BRSTD. Compared to HANet [81] (AP_{vt} = 10.9, AP_t = 22.2, and Params = 26.4), the top-performing remote sensing image model for very tiny object detection, BRSTD, attains 95.4% AP_{vt} of HANet with only 6.8% of its parameters. This highlights the efficiency and accuracy of our model design.

2) Experiment on VisDrone: For the selection of training and test sets in the Visdrone dataset, we adopted the most widely used approach: using the "train" set as the training set and the "val" set as the test set. The comparative experimental results of the testing performance are shown in Table II. To facilitate a detailed analysis and comparison of the results, we listed the input image sizes for each method and categorized these baseline methods into deep learning object detection methods and lightweight object detection methods.

From Table II, it can be seen that our model excels in parameter efficiency compared to other SOTA models. Specifically, the BRSTD model uses only 5.6% of the parameters of the lightweight HRFPN [94] (AP_{50} = 50.8 and Params = 32.1) while achieving 90.3% of its AP_{50} performance. Notably, the input image size for HRFPN is 1024 × 1024, which is twice the size of our input images. Compared to EMAattention [89] (AP = 30.4, AP_{50} = 49.7, and Params = 91.18), our BRSTD_l model, with the same input image size, surpasses it by 0.2 AP and 0.9 AP_{50} with only 6.9% of the parameters. In comparison to SDP [90] (AP = 30.2, AP_{50} = 52.5, and Params = 96.7), which uses an input size of 800 × 800, BRSTD_l achieves 96.4% of its AP_{50} value with just 6.5% of the parameters and, additionally, surpasses its AP value by 0.4. When compared to the biologically inspired small object detection model PRDet [15], despite PRDet's input image size being as large as 1644 × 1644, BRSTD achieves an AP value that is 6.2 points higher than that of PRDet.

TABLE II

COMPARISON WITH SOTA METHODS ON THE VISDRONE DATASET. RED REFERS TO THE BEST RESULTS AND BLUE REFERS TO THE SECOND-BEST RESULT. INDICATES THAT THE MODEL IS BIO-INSPIRED

Method	Input Size	AP	AP_{50}	Param/M
Deep learning object detection methods				
Faster RCNN[28]	-	17.2	31.0	41.2
Cascade RCNN[29]	-	22.6	38.8	69.0
DMNet[11]	1000*600	28.5	48.1	-
AMRNet[85]	1500*800	31.7	52.6	-
CDMNet[86]	1000*600	29.7	50.0	-
QueryDet[87]	-	28.3	48.1	-
NWD-RKA [79]	-	27.4	46.2	540.1
UAVOD[88]	-	24.4	47.6	-
EMA[89]	640*640	30.4	49.7	91.18
SDP[90]	800*800	30.2	52.5	96.7
PRDet † [15]	1644*1644	21.1	51.3	-
Light-weight object detection methods				
RetinaNet[91]	-	22.7	44.3	36.4
YOLOXs[92]	1536*1536	29.7	48.9	8.9
SR-YOLO[93]	-	23.9	41.6	7.61
YOLOv8-m[71]	640*640	25.2	41.8	25.8
HRFPN[94]	1024*1024	-	50.8	32.1
BRSTD † (our)	640*640	27.3	45.9	1.8
BRSTD_l † (our)	640*640	30.6	50.6	6.3

TABLE III

COMPARISON WITH OTHER STRONG BASELINES ON THE DOTA DATASET. NOTE THAT RED REFERS TO THE BEST RESULTS AND BLUE REFERS TO THE SECOND-BEST RESULT

Model	mAP50	Param/M	GFLOPs
Faster R-CNN[28]	60.6	60.2	118.8
RetainNet[91]	50.3	55.4	293.4
GFL[95]	66.5	19.1	159.2
FCOS[96]	67.7	31.6	202.2
ATSS[73]	66.8	18.9	156.0
MobileNetV2[97]	56.9	10.3	124.2
shuffleNet[98]	57.7	12.1	142.6
BRSTD(our)	65.4	1.8	64.3
BRSTD-l(our)	67.9	6.3	220.2

3) Experiment on DOTA: In the comparative experiments using the DOTA dataset, we utilized the "train" set as the training set and the "val" set as the test set. The results of the performance tests are presented in Table III. For evaluation metrics, we adopted the mAP50 index officially recommended by the DOTA dataset to assess the detection performance of the models.

In Table III, we compared two versions of our BRSTD model with other strong baselines on the DOTA dataset, including classic models in CV tasks [28], [73], [91], [95], [96]

TABLE IV

COMPARISON WITH DIFFERENT BASELINE YOLO FRAMEWORKS ON THE VISDRONE DATASET. NOTE THAT RED REFERS TO THE BEST RESULTS AND BLUE REFERS TO THE SECOND-BEST RESULT

Model	mAP50	Param/M	GFLOPs	FPS
YOLOv3-m[33]	44.5	47.9	122.1	94.9
YOLOv3-l	47.1	103.8	283.3	55.7
YOLOv5-m[34]	43.3	25.1	64.5	140.3
YOLOv5-l	45.6	53.2	135.6	97.1
YOLOv8-m[71]	44	25.8	78.7	138
YOLOv8-l	46.4	43.7	165.7	86.2
BRSTD(our)	47.7	1.8	64.3	143.5
BRSTD-l(our)	52.4	6.3	220.2	94.8

as well as renowned lightweight models [97], [98]. Overall, our BRSTD-I model, while reducing the parameter count by 25.3 M (87.6%), outperformed the second-highest detection performer, FCOS [96], by 0.2 mAP50 (0.3%). Compared to GFL [95], the model with the best overall performance among those compared, our model also achieved an improvement of 1.2 mAP50 (1.8%) and reduced the GFLOPs by 137.9 (68.2%). Our BRSTD model has a parameter count of only 1.8 M, which is substantially lower than the lightweight models MobileNetV2 [97] and ShuffleNet [98], with parameter reductions of 8.5 M (81%) and 10.3 M (85%), respectively, and computational reductions of 59.9 GFLOPs (48.2%) and 78.3 GFLOPs (54.9%). In addition, performance improvements of 8.5 mAP50 and 7.7 mAP50 fully demonstrate the advantages of the model we proposed.

4) Lightweight Experiment: To clearly demonstrate the advantages of our designed model in terms of lightweight performance, we conducted comparisons on the same device under identical experimental conditions. The BRSTD and BRSTD-l models were compared with various baseline YOLO series models [33], [34], [71] in terms of detection performance (mAP50), parameter count (Param/M), computational cost (GFLOPs), and running speed (FPS). The experimental results are recorded in Table IV.

As shown in Table IV, BRSTD achieved the best overall performance, with detection performance second only to our BRSTD-l. Compared to the YOLO series models included in the comparison, BRSTD's GFLOPs and FPS are superior to the lightest YOLOv5-m [34], and it achieved an increase of 4.4 in mAP50 with only 7.1% of its parameter count.

D. Ablation Studies

1) Experiment of XYWA and TDSA on AI-TOD: We first conducted ablation experiments on the AI-TOD dataset to investigate the impact of different components in BRSTD by evaluating the proposed XYWA and TDSA. Using the BRSTD network without XYWA and TDSA as the baseline, we studied the influence of various components on performance. As shown in Table V, we detailed the mAP50-95/mAP50 values for each category under different combinations of BRSTD, as well as the overall mAP50-95 and mAP50 values for all categories combined, along with the parameter count for each combination. Finally, we included the parameters of the enhanced model BRSTD_l for comparison to study the performance improvement with a slight increase in parameters.

As shown in Table V, compared to the baseline network, the addition of XYWA improves the mAP50-95 by 0.5 (0.9%) and mAP50 by 0.5 (2%), with only a 0.1 M increase in parameters. The addition of TDSA results in an mAP50-95 increase of 0.6 (2.5%) and mAP50 increase of 1.3 (2.3%), with a parameter increase of just 0.5 M. When both TDSA and XYWA are added simultaneously, the mAP50-95 increases by 1.8 (7.4%) and mAP50 by 2.9 (5.2%), with only a 0.6 M increase in parameters. This demonstrates that XYWA and TDSA can enhance the model's performance with minimal parameter increases, and their combined effect significantly outperforms the sum of their individual improvements.

In addition, compared to the model with both TDSA and XYWA modules, BRSTD_l increases mAP50-95 by 2.7 (10.4%) and mAP50 by 3.9 (6.6%) with only a 4.5 M increase in parameters. This indicates that the architecture of the enhanced model has a significant impact on effectively utilizing its additional capacity to improve detection accuracy.

2) Experiment of XYWA and TDSA on VisDrone: To demonstrate the generalizability of the proposed modules, we also conducted ablation experiments on the Visdrone dataset, following the same methodology as on the AI-TOD dataset. We used the BRSTD network without XYWA and TDSA as the baseline to study the impact of different components on performance. As shown in Table VI, we compared only the overall mAP50-95 and mAP50 for all categories on the Visdrone dataset and indicated the parameter count for each combination.

As shown in Table VI, in the ablation experiments on the VisDrone dataset, the results of adding XYWA or TDSA individually are significantly better than the baseline network, outperforming by 0.6 mAP50 and 1.1 mAP50, respectively, which fully demonstrates the effectiveness of the designed modules. Furthermore, the network with both XYWA and TDSA added performs better than the network with either module added individually, outperforming the baseline network by 1.2 mAP50-95 and 2.0 mAP50, demonstrating that the combined effect of the proposed modules is greater than the effect of using each module individually. In addition, BRSTD_l also shows a significant performance improvement.

3) Experiment of X, Y, and W Cell Pathway: To investigate the integrity of the proposed parallel pathways and the effectiveness of the simulated X, Y, and W cells, we conducted a detailed experimental analysis and evaluation of the proposed XYWConv on the VisDrone dataset. We sequentially replaced the three designed cell modules with 3×3 convolutional kernels, exploring the role of the X, Y, and W cells and the network performance under different pathway combinations. The experimental results are recorded in Table VII.

As shown in Table VII, the model with only the X pathway added outperforms the model without the cellular pathway by 1.4 mAP50, and the model with only the W pathway added exceeds the model without the cellular pathway by 0.7 mAP50. The model with only the Y pathway added shows the highest increase, outperforming by 2.0 mAP50-95 and 3.3 mAP50.

TABLE V
ABLATION STUDY OF XYWA AND TDSA ON THE AI_TOD DATASET

XYWA	TDSA	AI	BR	ST	SH	SP	VE	PE	WM	mAP 50-95	mAP 50	Param /M
		30.3/63.1	17.6/49.1	41.1/76.9	36.6/75.6	16.6/41.5	30.7/73.7	13.2/38.8	7.7/31.3	24.2	56.3	1.2
✔		29.8/63.5	19.2/50.8	42.1/78.4	36.4/75.5	19.1/45.7	30.8/74	13/38.4	7/28.2	24.7	56.8	1.3
	✔	29.4/61.7	17.8/49.8	41.1/78.2	38.6/77.8	17.4/45.2	31.6/74.8	13.7/40.3	8.4/32.8	24.8	57.6	1.7
✔	✔	35.6/73	17.2/48.7	43.1/80	38.6/77.3	18.7/46	32.3/75.5	14.6/41.6	7/31.2	26	59.2	1.8
BVANet-l		38.2/73.1	21.1/53.4	44.4/82.2	42/81.2	20.6/47.6	34.2/77.5	17.4/48	11.3/41.5	28.7	63.1	6.3

TABLE VI
ABLATION STUDY OF XYWA AND TDSA ON THE VISDRONE DATASET

XYWA	TDSA	mAP 50-95	mAP 50	Param /M
		27.7	45.7	1.2
✔		28.2	46.3	1.3
	✔	28.5	46.8	1.7
✔	✔	28.9	47.7	1.8
BVANet-l		32.3	52.4	6.3

TABLE VII
ABLATION STUDY OF X, Y, AND W CELL CONVS

X	Y	W	mAP50-95	mAP50
✗	✗	✗	25.9	43.0
✔	✗	✗	26.7	44.4
✗	✔	✗	27.9	46.3
✗	✗	✔	26.4	43.7
✔	✔	✗	28.8	47.5
✔	✗	✔	26.9	44.8
✗	✔	✔	28.5	46.8
✔	✔	✔	28.9	47.7

This fully demonstrates the effectiveness of each individual cellular pathway. Furthermore, any combination of two cell pathways in parallel performs better than a single-cell pathway. When the X, Y, and W cell pathways form a complete parallel pathway, the model achieves the best performance, reaching 28.9 mAP50-95 and 47.7 mAP50, which are, respectively, 3.0 and 4.7 higher than the model with the cellular pathway. This indicates that as the parallel pathways are gradually constructed, the overall performance of the model shows an upward trend.

E. Experiment Analysis

To visually demonstrate the detection performance of our proposed model in different scenarios, we compared the detection results of BRSTD with those of YOLOv8-m and ground truth, as shown in Fig. 8.

In Fig. 8(a), we mainly focus on the detection ability for very tiny objects. It can be observed that YOLOv8-m misses several tiny objects, and although it detects some, the results are not as satisfactory compared to ground truth. In contrast, BRSTD significantly improves the detection capability for tiny objects, with results highly consistent with ground truth, demonstrating strong detection ability for tiny objects.

In Fig. 8(b), there are many complex background elements around the ships docked at the pier. While YOLOv8-m can detect objects in complex backgrounds, it still misses many. Compared to YOLOv8-m, BRSTD shows markedly better detection results in complex backgrounds. However, in comparison with ground truth, there are two false positives in the red magnified box at the top. Upon enlarging the original image, it is difficult even for the human eye to not classify them as ships, especially the object in the top-right corner.

In Fig. 8(c), we analyzed the performance of the two algorithms in detecting densely arranged objects. YOLOv8-m fails to detect densely stacked bicycles, whereas BRSTD performs much better in handling dense objects, with results nearing those of ground truth.

In Fig. 8(d), we examined the detection performance for multicategory objects with large size variations. Both YOLOv8-m and BRSTD perform well in detecting medium to large objects; however, YOLOv8-m encounters significant detection failures with distant tiny objects. Although BRSTD has some false negatives, it shows considerable improvement over YOLOv8-m.

Through experimental comparative analysis of YOLOv8-m and BRSTD in different detection scenarios, BRSTD excels in detecting very tiny objects, handling complex backgrounds, detecting dense objects, and identifying tiny objects across multiple categories and size variations, significantly outperforming YOLOv8-m. This indicates that the BRSTD algorithm is more robust and accurate in various challenging detection tasks, proving to be an effective method for detecting tiny objects.

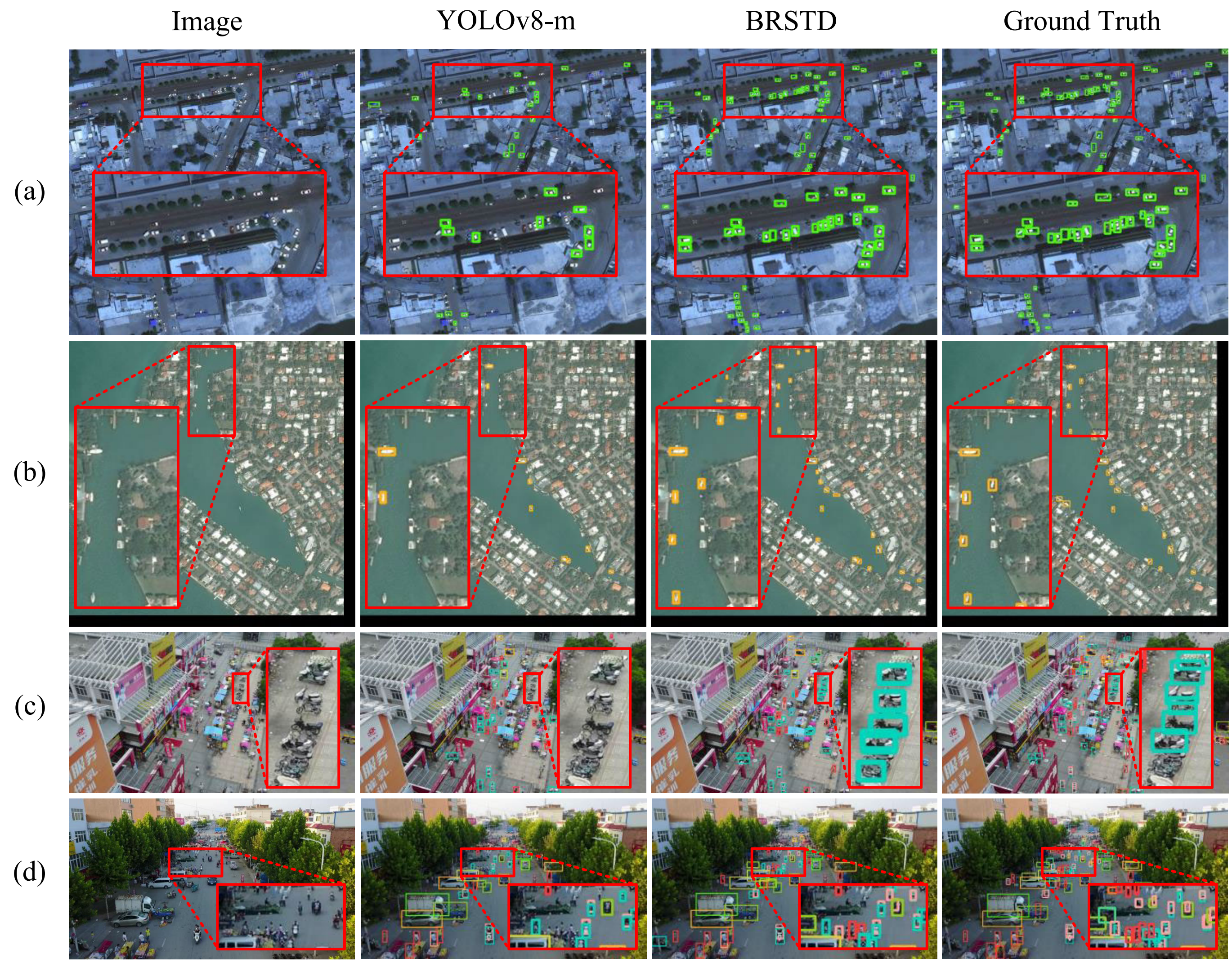


Fig. 8. Comparison of YOLOv8-m and BRSTD output results with ground truth. (a) Detection of very tiny objects. (b) Detection under complex background information. (c) Detection of dense objects. (d) Detection of multiple categories of objects with large size differences.

V. Conclusion

Inspired by biological vision, this article new attention-guided tiny object detection model. Drawing inspiration from the parallel pathways of visual perception and the receptive field characteristics of XYW cells, we developed a new XYWConv module. Building on this, we designed a new spatial channel attention mechanism, XYWA, inspired by the bottom–up attention mechanism in biological vision, which aligns with the V1 saliency model and the Mexican hat model, forming a new backbone structure. At the junction of the neck and head of the model, inspired by the top–down attention in biological vision, we designed a feedback inhibition attention module, TDSA, that uses deep information to modulate shallow information, thereby better preserving information about tiny objects in shallow feature layers while suppressing information about larger objects and background. Through ablation studies on three widely used tiny object datasets, we have proven the effectiveness of our designed modules. Compared to other advanced object detection models, our approach achieves superior performance even with significantly reduced parameters.

However, this article still has several limitations. For instance, the use of multiscale convolutional kernels, dilated convolutions, and parallel convolutions to simulate the receptive fields of cells has achieved good detection performance, but it also introduces considerable computational complexity. Moreover, the research in this article focuses solely on the bionic simulation from the perspective of parallel pathway cell receptive fields and attention, without addressing the brain's visual processing mechanisms. In future work, we will further study and improve the retina/LGN-V1 parallel pathways, directly simulating the receptive fields on convolutional kernels to avoid extensive high-complexity calculations. In addition, we will focus on the "What" pathway, which is responsible for object recognition and visual feature analysis in the visual system. We plan to conduct bionic simulations of the receptive fields for each visual area (V1–V2–V4) within the "What" pathway. Furthermore, we will explore the bionic simulation of the attention mechanisms involved in top–down modulation between layers of the "What" pathway and design a novel feedback-fusion attention module. This future work is expected to achieve superior performance with lower computational complexity.

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